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This paper has been accepted for publication in the *2025 IEEE Intelligent Vehicles Symposium (IV)*, held in Reykjavík, Iceland.

Final version available at: `https://ieeexplore.ieee.org/`

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Randomized Model Predictive Control for Autonomous Racing

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***Abstract*—Autonomous racing is an emerging research field that has gained significant interest in recent years. Controlling a vehicle at the limit of tire grip presents a challenging control problem. This paper demonstrates the implementation of a randomized optimization based Model Predictive Control (MPC), referred to as Randomized Model Predictive Control (RMPC), to address the challenge of autonomous racing. The random samples in the proposed RMPC are generated using a novel generation technique in the frequency domain. This approach prevents undesirable oscillations, which are critical to the smoothness required for vehicle control. The proposed RMPC is implemented on a Graphics Processing Unit (GPU), minimizing data transfer to the Central Processing Unit (CPU). The results demonstrate improved control performance and lap times that surpass those achievable with state-of-the-art nonlinear control methods. The proposed scheme proved fast and efficient in driving a full-scale autonomous race car built on a Dallara SF23 Superformula chassis. As part of the Abu Dhabi Autonomous Racing League (A2RL) race of 2024, the proposed implementation set the fastest autonomous lap of the series at the iconic Yas Marina Circuit, achieving a lap time within 10% of a Formula 1 driver in the human-driven version of the same car.**

***Index Terms*—Autonomous vehicles, model predictive control, sampling based optimization, Graphics Processing Unit (GPU).**

I. Introduction

AUTONOMOUS racing has been attracting increased attention in recent years. The emergence of races such as Roborace, the Indy Autonomous Challenge, and A2RL [1] serves as evidence of this growing interest. The development of advanced controllers for autonomous racing is a critical area of research, bridging state-of-the-art control theory, system identification, and machine learning to address the unique challenges of high-speed dynamic environments. Autonomous racing serves as a testing ground for algorithms designed to achieve optimal performance under constraints such as limited traction, rapid decision-making, and the need for robust adaptability to unpredictable scenarios.

At its core, the goal of autonomous racing is to maximize performance metrics such as lap time, stability, and precision while navigating a competitive track. This requires controllers capable of handling non-linear dynamics, time-varying parameters, and interactions with other vehicles or environmental changes. The challenges in autonomous racing are not just theoretical; real-world implementations must address sensor noise, latency, and uncertainties. Traditional model-based approaches such as Model Predictive Control (MPC) have been widely adopted due to their ability to handle constraints and optimize control inputs over a finite horizon [2]–[5]. However, these methods can face computational challenges at high speeds when they use precision models. This necessitates innovations in real-time optimization techniques and hardware acceleration.



Fig. 1: EAV24 race car of Technology Innovation Institute (TII) at A2RL 2024

Recently, learning-based methods have also gained traction in autonomous racing [6]–[8]. Reinforcement Learning (RL), for instance, enables agents to learn optimal driving strategies through interaction with the environment. Simulation studies have demonstrated the integration of RL with model-based controllers to achieve better generalization and adaptability [9]. Furthermore, neural network-based approaches have been employed to approximate vehicle dynamics or directly map sensor inputs to control outputs, highlighting the potential of data-driven techniques to complement traditional methods [8], [10]. This class of methods are demonstrated only in simulations. They are yet to be applied to fast autonomous driving of full scale cars.

RMPC is a type of MPC that was demonstrated by the authors previous works to be extremely fast and computationally

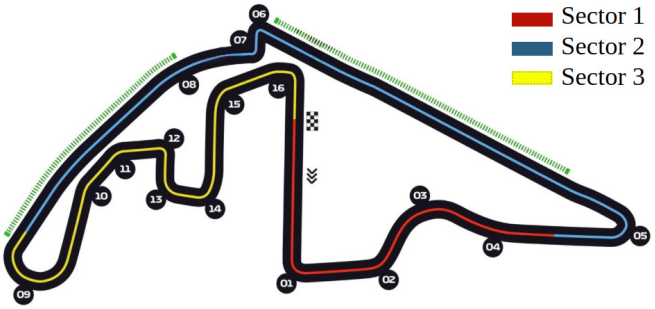


Fig. 2: Sectors and corners in Yas Marina F1 circuit (YMC), Abu Dhabi

efficient while controlling complex non-linear systems without approximations. This concept was successfully used for fast obstacle avoidance driving [11]–[13] of RC cars. RMPC was also used in a stochastic MPC framework to autonomous driving around pedestrians and other high entropy road users [14], [15]. This paper aims to introduce a novel controller designed for autonomous racing, leveraging the strengths of RMPC and its implementation in GPU. By addressing the computational and practical challenges of real-time operation, the proposed method contributes to advancing the field of autonomous racing, with implications for broader applications in autonomous driving and robotics.

In summary, This work has the following major contributions:

- Implementation of a random sampling based model predictive control for multiple control inputs for autonomous driving at close to human speeds.
- Demonstrates a fully parallel implementation on GPU realizing fast computational times with minimum CPU utilization.
- Achieves lap times that are within 10% of a Formula one driver in a Formula one circuit.

As the second season of A2RL approaches, the authors and other competitors remain dedicated to the pursuit of achieving superhuman lap times and enabling intelligent, close multi-car racing with autonomous race cars. However, the realization of superhuman lap performance continues to present challenges, primarily due to the extraordinary human capacity to perceive and operate at the limits of a vehicle's performance envelope.

II. Problem Description

The target problem in autonomous racing is a lap time minimization problem around a given circuit. The car tries to follow a reference race line that includes position and speed references while guaranteeing the stability of the car. In this work and in the initial race, we consider qualifying style laps with no opponents.

The optimum reference trajectories are generated by a combination of two race lines. The first is a mathematically optimum race line, and the second is the race line of a professional driver. The professional driver race line was recorded using a human-driven version of the car with the same chassis and powertrain. The optimum trajectory generation problem takes the position and velocity of the driver as a starting point to generate a smooth reference that obeys the system constraints. The readers are referred to [16] for the details of reference line generation.

A planner module provides the optimal race line as position and velocities to the controller. This module publishes a chunk of the race line starting at the current position of the ego car. This trajectory lasts until a fixed time in the future, discretized by a time step, providing the controller with N_p points. This reference trajectory is represented as follows:

$$
\begin{aligned}
{}^{l}\chi_{ref} &= \{{}^{l}p_{ref,i}, {}^{l}v_{ref,i} | i = 0, 1, \cdots, N_p\}, \\
{}^{l}p_{ref,i} &= \{{}^{l}x^{p}_{ref,i}, {}^{l}y^{p}_{ref,i}, {}^{l}\theta_{ref,i}\}, \\
{}^{l}v_{ref,i} &= \{{}^{l}x^{v}_{ref,i}, {}^{l}y^{v}_{ref,i}, {}^{l}r_{ref,i}\},
\end{aligned} \tag{1}
$$

where ${}^{l}\chi_{ref}$ is the reference path at time t that contains N_p path nodes. ${}^{l}p_{ref,i}$ denotes the relative position of the path node, ${}^{l}x^{p}_{ref,i}$ and ${}^{l}y^{p}_{ref,i}$, from the car's origin and ${}^{l}\theta_{ref,i}$ is the relative heading. ${}^{l}v_{ref,i}$ represents the velocity ${}^{l}x^{v}_{ref,i}$ and ${}^{l}y^{v}_{ref,i}$ at the respective point on the path, and ${}^{l}r_{ref,i}$ is the yaw rate.

The sectors and corners of Yas Marina Circuit can be found in Fig. 2. The lap times car counted starting from the start finish line indicated by the chequered flag in Fig. 2. The given task of achieving the fastest possible lap time involves optimizing the vehicle's trajectory, speed, and control inputs to minimize lap time. The main goal is to balance speed, grip, and control to execute the most efficient line around the track. This is achieved through precise braking, optimal cornering, and effective acceleration out of turns. The ideal approach takes into account the physics of the car (such as weight transfer, aerodynamic downforce, and tire grip) as well as circuit characteristics (track surface, elevation changes, and corner radius) [17].

In this particular race track, there are certain corners that are more challenging than others. Turn 6 and 7 present a tight chicane after the longest straight of the track. Entry to turn 6 presents the heaviest braking followed by very high steering angles. A hotel building covers the straight between turns 13 and 14, causing temporary GPS degradation and often leading to sudden steering corrections immediately after turn 14. Turn 9 is particularly noteworthy due to constraints on lateral grip. Precise throttle control is required because the tires operate near their lateral force saturation throughout this turn. Similarly, turn 12 warrants attention as it features a relatively high entry speed compared to the rest of the circuit.

III. Vehicle Model

The EAV24 vehicle used for the A2RL event is depicted in Figure 1. It is is based on the Dallara SF23 chassis, which is used in the Japanese Super Formula Championship. The powertrain consists of a four-cylinder turbocharged engine and the vehicle is fitted with Yokohama Super Formula tires.

A. Curvilinear Single Track Model

A single track dynamic model, shown in Figure 3, is used for the Nonlinear Model Predictive Control (NMPC). As in [18], a curvilinear/Frenét coordinate frame is used to describe the state. Therefore, the global position and heading are not directly considered, but transformed to a state relative to the reference path. The vehicle model used in this work is an extension of the model used in [19].

1) Equations of Motion: The vehicle state and the input are given by

$$x = \begin{bmatrix} s & y_e & \theta_e & v_x & v_y & r & \delta & T & B & G \end{bmatrix}^T \quad (2)$$

$$u = \begin{bmatrix} \delta & T & B & G \end{bmatrix}^T . \quad (3)$$

The variables s, y_e, and θ_e denote the progress along the path, the orthogonal deviation from the path, and the local heading, respectively. The longitudinal and lateral velocities $\dot{s}$ and $\dot{y_e}$ are considered, along with the yaw rate $\dot{\theta}_e$. The steering angle δ, throttle command T, brake command B, and gear command G represent the control inputs.

Thus, the equations of motion are

$$\begin{aligned} \dot{s} &= \frac{v_x \cos(\theta_e) - v_y \sin(\theta_e)}{1 - y_e \rho_{ref}(s)} , \\ \dot{y_e} &= v_x \sin(\theta_e) + v_y \cos(\theta_e) , \\ \dot{\theta}_e &= r - \rho_{ref}(s)\dot{s} , \\ \dot{v_x} &= \frac{1}{m}\left(F_{x_r} - F_d - F_{y_f}\sin(\delta) + F_{x_f}\cos(\delta) + m v_y r\right) , \\ \dot{v_y} &= \frac{1}{m}\left(F_{y_r} + F_{y_f}\cos(\delta) + F_{x_f}\sin(\delta) - m v_x r\right) , \\ \dot{r} &= \frac{1}{I_z}\left(l_f\left(F_{y_f}\cos(\delta) + F_{x_f}\sin(\delta)\right) - l_r F_{y_r}\right) , \end{aligned} \quad (4)$$

where $\rho_{ref}(s) = r_{ref} \,/x^v_{ref}$ is the curvature at the progress

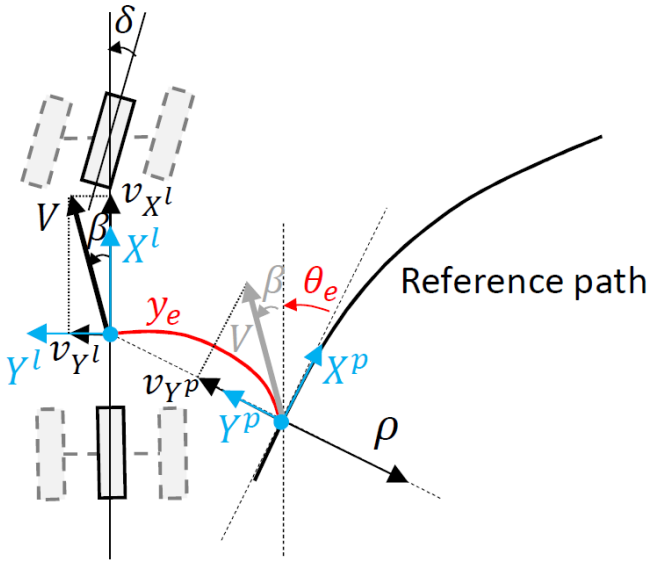


Fig. 3: Dynamic single track vehicle model on curvilinear coordinates.

s, l_f and l_r are the distances from the center of gravity to the front and rear wheels, m is the mass and I_z the moment of inertia. F_{y_f}, F_{y_r} are the lateral tire forces at the front and rear wheels. F_{x_f}, F_{x_r} are the longitudinal forces at front and rear axles. Forces due to track banking and gradient are minimal and not considered. F_d represents the aerodynamic force modelled as,

$$F_d = 0.5\,\rho_{air}\,A\,C_d\,v_x^2 . \quad (5)$$

The air density ρ_{air}, the frontal area A and the drag coefficient C_d are used in (5).

2) Tire Model: The tires effects are modeled using a simplified Pacejka Magic Formula [20]. Beyond the usual macro-parameters B, C, D and E, the lateral force offsets Sv_{yi}, $i \in [f, r]$, have been included resulting in

$$\begin{aligned} F_{y_f,\text{lat}} &= Sv_{y_f} + D_f \sin\Big(C_f \tan^{-1}(B_f\,\alpha_f) + \\ &\quad - E_f\big(B_f\,\alpha_f - \tan^{-1}(B_f\,\alpha_f)\big)\Big) , \\ F_{y_r,\text{lat}} &= Sv_{y_r} + D_r \sin\Big(C_r \tan^{-1}(B_r\,\alpha_r) + \\ &\quad - E_r\big(B_r\,\alpha_r - \tan^{-1}(B_r\,\alpha_r)\big)\Big) , \end{aligned} \quad (6)$$

where α_i, $i \in [f, r]$, is the slip angle at the front and the rear calculated by

$$\begin{aligned} \alpha_f &= \tan^{-1}\left(\frac{v_y + l_f\,r}{v_x}\right) - \delta , \\ \alpha_r &= \tan^{-1}\left(\frac{v_y - l_r\,r}{v_x}\right) . \end{aligned}$$

3) Longitudinal Forces: The front axle longitudinal forces are modeled as

$$F_{x_f} = -C_{b_f} B - C_{ro_f} ,$$

where C_{ro_f} is the rolling resistance at front axle. The braking force is represented as $C_{b_f} B$, with C_{b_f} the maximum brake force and $B \in [0, 1]$.

Considering a rear wheels drive layout, the rear longitudinal force is modeled as

$$F_{x_r} = F_{engine}(T, v_x, G) - C_{b_r} B - C_{ro_r} , \quad (7)$$

where $F_{engine}(T, v_x, G)$ is a non linear function representing the drive force from the engine at the rear wheels. The turbo-charged combustion engine used in EAV24 produces a force which is non linear in the usable regions since it depends on the engine rpm and gear. In order to represent this behavior, the function (7) of engine torque is modeled on engine bench data, later refined with test data from the car. Gear G is considered as a state and is considered to be constant in the prediction horizon. The effects of shifting are neglected and the gear command is handled by a separate logic that request a shift to the low level when reaching the desired engine rpm.

B. Model Identification

The initial models were identified from the specification data for the car, engine and tires. A Proportional-Integral-Derivative (PID) controller with feed forward components based on [21], [22] was then used to perform simple driving maneuvers until speeds of 30 m/s. This data was used to identify and refine the models further for the MPC.

1) Tire Model Identification: The tires used for A2RL are Yokohama Advan tires, same as Japanese Super Formula championship's SF23 car. The initial tire model was provided by the tire maker using a test rig. It has been observed that the model could not be used to accurately reproduce the real tire behavior. The real behavior of the tires are highly affected by the thermally sensitive tire-road grip, wear, and suspension setup. The tire model based on the manufacturer was then refined with data collected from the race track. However, the proposed framework assumed a constant maximum tire grip, which became a limiting factor during tire warm-up. To mitigate this, a conservative grip value was used in the controller until the tires reached the appropriate operating temperature.

The figure Fig. 4 shows the estimated pacejka model on the force data from the car. The forces are estimated following the vehicle model using inputs from the state estimation of the car.

IV. Controller

The controller used in this work is an MPC with a randomized optimization solver with the necessary constraints. In order to represent the dynamics of the car, a dynamic bicycle model (refer section III) is used. Many of the previous works that implemented MPC for navigation [23] have adopted a point-mass model to simplify the problem. Due to the highly non linear forces that comes into play at high speed driving, a dynamic bicycle model is of choice. To keep the vehicle within its stability limits while driving very close to it, this framework utilizes some input and safety constraints, as shown in section IV-A.

For reliable real-time performance, a sample-based approach is chosen to solve the nonlinear optimization problem. The sample-based method used in this work is known as RMPC, and is attracting increased attention recently [11]–[15]. The algorithm 1 explains the steps in the RMPC process. Random sampling based methods often lead to higher noise levels in the control input. This is not recommended for smoothness critical tasks such as driving. The authors previous work [11] presents a method of sampling from the frequency domain called Inverse Discrete Cosine Transform (IDCT). This method has shown to provide significant improvement in smoothness of control inputs. In [13], the IDCT was implemented in a GPU for obstacle avoidance driving of an RC car. The readers are referred to our previous work [12] for further implementation details.

The previous works on RMPC were all limited to single output systems. This work extends it to smooth control of multiple inputs, namely throttle, brakes and steering. Another notable contribution of this work is that the RMPC is applied to drive a full scale race car.

Algorithm 1 Randomized MPC

```
for Every time step do
    Generate Ns steering samples and Ns force samples of length N satisfying (8) [11].
    Initialize state matrix.
    for Sample = 1, 2, ..., Ns^2 do
        for Length = 1, 2, ..., N do
            Calculate the extended state matrix acc. to (4).
        end for
    end for
    for Sample = 1, 2, ..., Ns^2 do
        for Length = 1, 2, ..., N do
            Calculate the cost for each sample acc. to (11).
        end for
    end for
    Find minimum cost sample Smin.
    Return second element of Smin as next control input.
end for
```

A. Input and safety constraints

Two types of constraints are used in this formulation. The first are the hard constraints that ensure the feasibility of input commands. These constraints represent the operation range of the hardware at the given time. The second type are the soft constraints that keeps the vehicle within it's dynamic limits.

The range of control inputs u and their rate of change $\dot{u}$ are constrained as follows to maintain physical limits of actuation systems and to prevent jerky inputs, respectively.

$$\begin{aligned} -100\% < \delta <\ & 100\%, \\ 0\% < T <\ & 100, \% \\ 0\% < B <\ & 100, \% \\ 0 < G <\ & 6, \end{aligned} \tag{8}$$

where, δ is the steering angle measured at the wheel. 100% is 20 degrees at the wheel. Throttle T ranges from 0 to 100% of the actuator. For brakes B, 100% indicates 120 bars of brake pressure at the front axle, this is then scaled at a fixed brake bias and applied to the rear axle. Gear G ranges from 0 to 6, where 0 is neutral.

The $\dot{u}$ constraints are

$$\begin{aligned} -100\%/s < \dot{\delta} <\ & 100\%/s, \\ -600\%/s < \dot{T} <\ & 600\%/s, \\ -550\%/s\% < \dot{B} <\ & 550\%/s. \end{aligned} \tag{9}$$

While the car is driving fast, it is kept stable by making sure the tires are not pushed beyond their maximum grip limit. The maximum combined force is limited at the tire contact patch

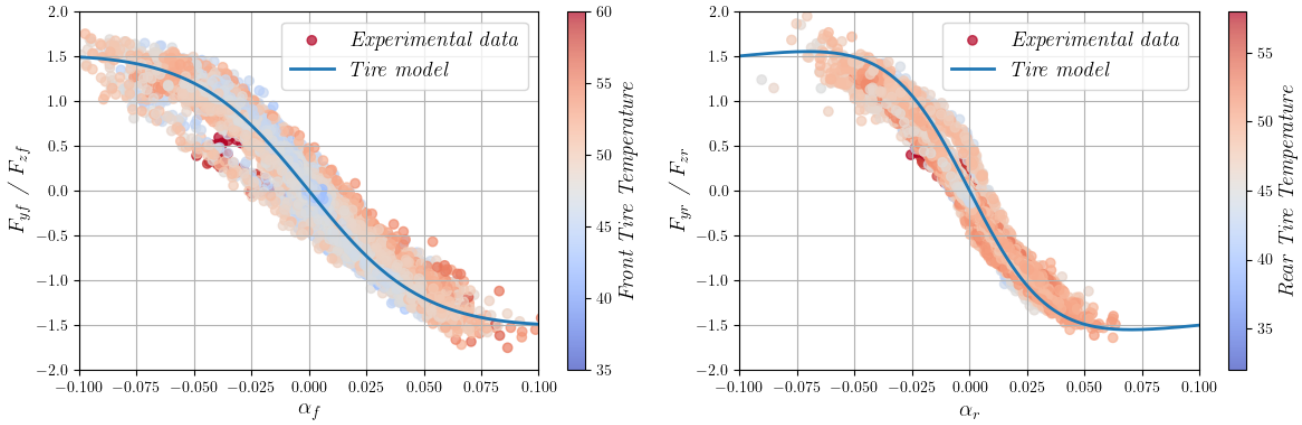


Fig. 4: The estimated tire model over experimental data gathered during A2RL

with a soft constraint added to the cost. The combined force is constrained as follows.

$$F_i = \sqrt{(F_{y_i})^2 + (F_{x_i})^2} \leq \mu_i F_{z_i}, \quad \forall i \in \{f, r\}. \tag{10}$$

F_i represents the total force at the tire i. Forces F_{x_i}, F_{y_i} and F_{z_i} are the forces along x, y and z directions, respectively. Tyre index i can be f or r representing front and rear axles. Please note that both left and right tires are considered as one combined tire since a bicycle model is employed. μ represents the friction coefficient between the tire on the road.

B. Cost function and reference state

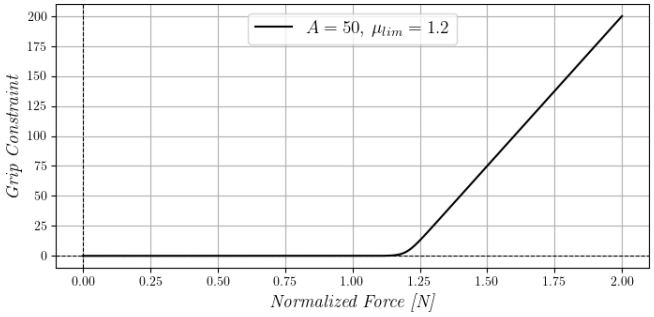


Fig. 5: Shape of the force constraint

The controller aims to track the reference trajectory as closely as possible without exceeding the vehicle's dynamic limits. In practice, these objectives may conflict when higher speeds require using a wider lane or slowing down to avoid loss of control and prevent spin-outs. This trade-off is addressed by incorporating a grip constraint into the cost function. Assigning different weights to various cost terms balances fidelity to the reference path with adherence to stability constraints. Finally, additional penalties on the control inputs ensure smooth actuation.

The cost function $J(u_t)$ for an input series $u_t = [u(0|t), u(1|t), \cdots, u(N-1|t)]$ to be optimized at time t is

$$\begin{aligned} J(u_t) = &\sum_{k=1}^{N-1} \left(\phi^T(k|t) Q \phi(k|t) + \Delta u(k|t)^T R \Delta u(k|t)\right) \\ &+ \phi^T(N|t) Q_f \phi(N|t) \\ &+ Q^f_{grip} \sum_{k=1}^{N-1} P^f_{grip}(k|t) + Q^r_{grip} \sum_{k=1}^{N-1} P^r_{grip}(k|t) \end{aligned} \tag{11}$$

$$\text{where} \quad \phi(k|t) = y(k|t), \tag{12}$$

$$\Delta u(0|t) = 0, \ \Delta u(k|t) = |u(k|t) - u(k-1|t)|_1 \quad \forall k \in \{1, 2, \cdots, N-1\}, \tag{13}$$

where N, ϕ and Δu represents prediction horizon length of MPC, state error and the time difference in control input, respectively. The representation $u(k|t)$ is the value of u at prediction horizon step k at time t. $Q = diag(Q_{y_e}, Q_{\theta_e})$ and $R = diag(R_{y_e}, R_{\theta_e})$ being weight parameters, balance the control performance to the effort. $Q^f = diag(Q^f_{y_e}, Q^f_{\theta_e})$ acts on the last step in the prediction horizon as a penalty to the residue.

This framework enables the controller to follow any reference path χ^{ref}. The reader can refer to our previous work [24] for a detailed study on the path tracking performance of this dynamic model. Further details on the weight parameter matrices can be found in author's previous work [13].

Q^f_{grip} and Q^r_{grip} represent the weight parameters for the grip constraint. The cost when front or rear axle is approaching the grip limit μ_{lim} is expressed by P^j_{grip}:

$$P^i_{grip}(k|t) = \left(\log\left(1 + \exp\left(A \cdot \left(\frac{F_i(k|t)}{f^i_z(k|t)} - \mu_{lim}\right)\right)\right) - \log\left(1 + \exp\left(-A \cdot \mu_{lim}\right)\right) \right) \tag{14}$$

where $F_i(k|t)$ and $f^i_z(k|t)$ are the total force and vertical force at axle i predicted for prediction horizon step k at time t. The parameter A can adjust the shape of the grip limit constraint, $i \in \{f, r\}$ indicates the front and rear axle respectively. The shape of the constraint in use is as as shown in fig 5.

C. Formulation of input optimization problem

The optimization problem to be solved in every control step in order to generate optimum control input series can be formulated as follows:

given

$$x(0|t) = x(t),\ \chi^{ref}, \tag{15}$$

find

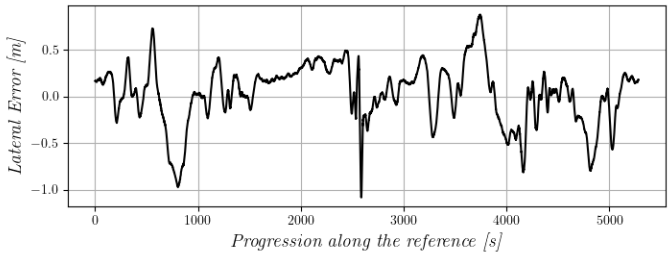
$$u(k|t), \qquad (k \in \{0, 1, \cdots, N-1\}) \tag{16}$$

which minimize

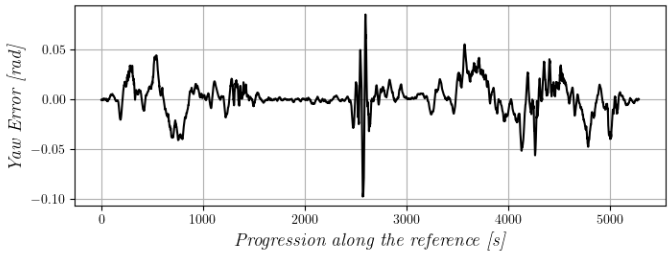
$$J(u_t = \{u(k|t)\}), \qquad (k \in \{0, 1, 2, \cdots, N-1\}) \tag{17}$$

subject to

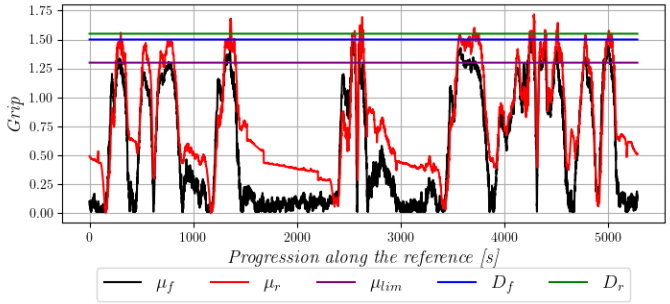
$$\begin{aligned} &\text{Input constraints} \quad (8), (9) \\ &\text{Grip limit constraints} \quad (14) \\ &\text{Car dynamics} \quad (4). \end{aligned} \tag{18}$$



(a) Lateral error



(b) Yaw error



(c) Grip

Fig. 6: Lateral tracking

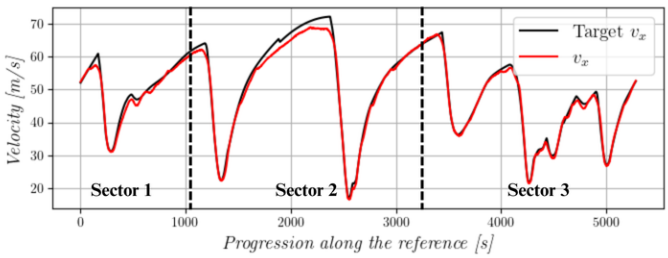
V. Results

The target of the controller is to track the trajectory sent by the planner as close as possible while not surpassing the vehicle dynamics limits of the car. All the plots in this section are from our fastest lap of 115.72 (s) in the 2024 A2RL race. On the x-axis, s is used to denote the progress along the reference racing line.

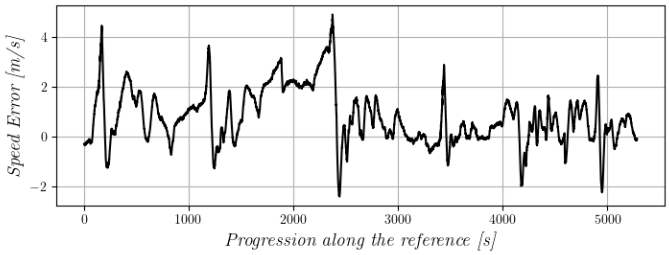
The figure Fig.6 shows the tracking performance of the controller in terms of lateral and yaw errors from the reference. The reference trajectory used is designed with at least 1 (m) margin from the curbs of the race track. The controller is able to stay within the 1 (m) of lateral error allowed by reference. The turns with the highest lateral error are the turns 4,6 and 11. These are at progression 800, 2600 and 3750, respectively. As seen in Fig.6(c), these are places where the car is also close to the grip limit, hence the controller relaxes the reference tracking in order to favor stability.

On the other hand, Fig.7 shows the longitudinal tracking performance of the controller. The speed tracking error is within +/- 4 (m/s). Carrying speeds higher than the reference into a corner was found to compromise lap times by compromising tracking and stability. At it's limits this can also cause uncontrolled spins. The controller tuning for speed tracking was mostly focused on ensuring there is no speed overshoots while taking corners. Figure Fig.7 shows turn 6 at progression 2500. Turn 6 following the longest straight is the heaviest braking zone with the highest speed over shoot in the whole lap. It can be seen that the controller begins to steer only after the speed over shoot of 4 (m/s) is rectified. This saves the car from spinning out. Two reasons were identified for the speed error to exhibit oscillations. The first reason is the absence of gear shifts in the prediction horizon. This triggers throttle oscillation after some gear shifts as seen in Fig. 8a. The second reason is the inaccuracies in the engine torque model. Since the engine is turbocharged, it proved challenging to capture all the dynamics in a simple enough model for the controller to handle in real time. These two factors will be addressed in our future work.

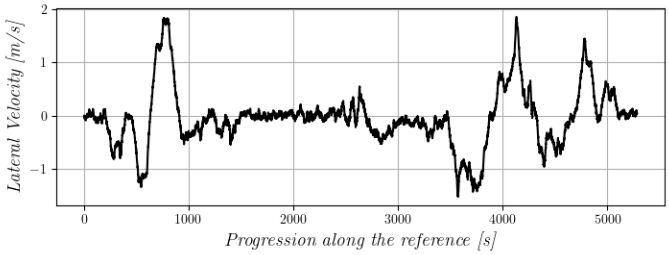
There are several signals that are analyzed to assess the stability of the car at high speeds. Figure Fig.8 illustrates the



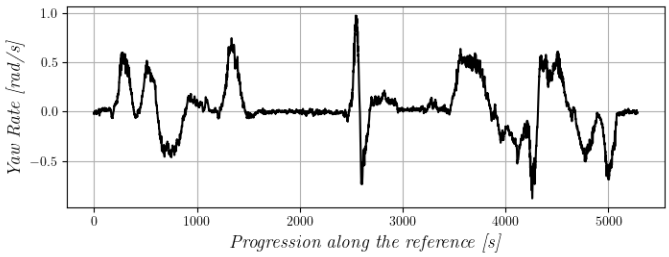
(a) Longitudinal velocity



(b) Speed error

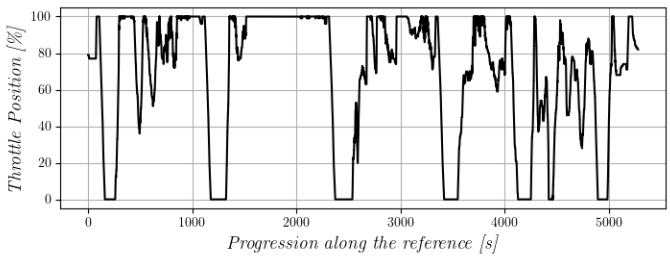


(c) Lateral velocity

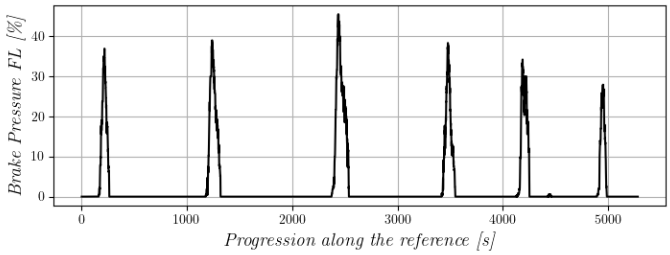


(d) Yaw rate

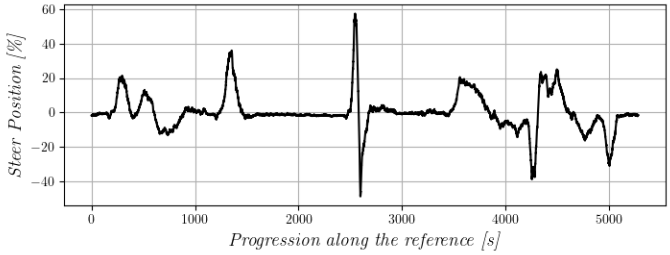
Fig. 7: Speed tracking



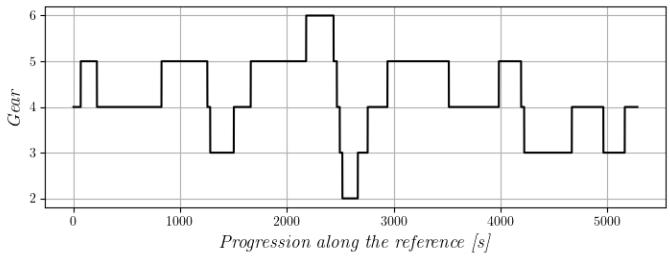
(a) Throttle input



(b) Brake input



(c) Steering input



(d) Gear input

Fig. 8: Control inputs for the system

control inputs to the car. The throttle application at turn exits and the brake signal profiles closely resemble those of a human driver, thus appearing natural.

The Fig.7 also illustrates the car speed, along with the control inputs and the yaw rate. Sudden spikes in the yaw rate clearly indicate instability. By analyzing human driving telemetry and extensive testing, we have determined limiting trends in yaw rate that indicates the dynamic limits of the car. The controller constraints are then adjusted to ensure that the yaw rate remains within these limits.

VI. Conclusion

The proposed RMPC is demonstrated to realize tight dynamics-aware trajectory tracking while driving a super formula car within it's vehicle dynamic limits in a formula one circuit. The controller based on randomized optimization is able to simultaneously control both lateral and longitudinal inputs to the car.

The on-board computer is a neousys RGS-8805GC with an AMD EPYC™ 7003 series "MILAN" CPU and NVIDIA® RTX A6000 GPU. This controller achieves a computational time of 3ms on the given computer. A control frequency of 100 Hz is employed, leveraging the fast computation enabled

TABLE I: Details lap times

	Lap time (s)			
Car	Sector 1	Sector 2	Sector 3	Total
Human car	22.4	46.7	36.7	105.34
TII	24.57	48.43	42.73	115.72
POLIMOVE	24.35	48.75	44.75	117.85
TUM	25.13	48.96	44.89	118.99

by the GPU.

A human-driven version of the same chassis was prepared for the A2RL event of April 2024. The two cars are identical, except for an increased weight by 32 (kg) of the autonomous. Assuming a 60 (kg) driver and a substitution of the driver with the autonomous stack, there is also a shift in center of gravity by 0.21 (m) to the front and 0.07 (m) upwards respectively. The human version driven by ex-Formula One driver Daniil Kvyat achieved a lap times of 105.34 (s) during the event. The autonomous version driven by the proposed controller was able to achieve lap times of 115.72 (s), leaving the gap to a professional human driver below 10%. Please refer to Table I for detailed sector times. The sector and lap times of the top 3 autonomous teams in A2RL 2024 are taken from the official timekeeping of the race [25].

The extension of this work is focused on achieving faster than human lap times. The addition of gear shifts as one of the control commands will be explored. Gear shift being a non continuous signal, is currently handled external to the RMPC. Integration of a neural network trained on track data to improve the model accuracy is also in the scope of future work.

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